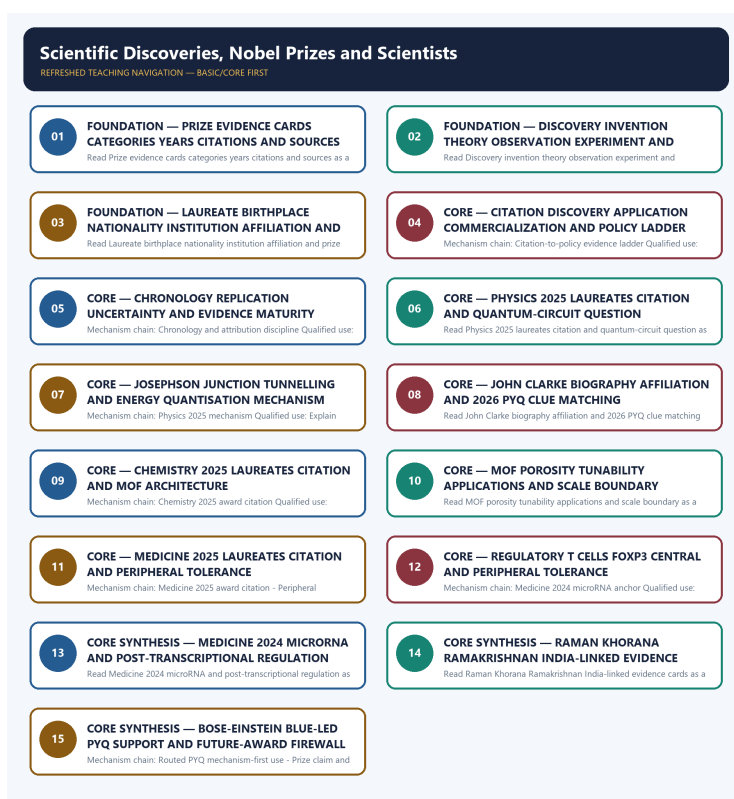


## SOLVED PRACTICE WORKBOOK

# Scientific Discoveries, Nobel Prizes and Scientists - Solved Practice Workbook

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing



Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

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# Scientific Discoveries, Nobel Prizes and Scientists - Solved Practice Workbook

## BASIC MCQS / REMEDIATION

### Q1. Which statement correctly identifies Minimum prize evidence card?

A. A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. B. A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. C. The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. D. Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution.

**Answer: A. Explanation: A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q2. Which option preserves the technical boundary of Minimum prize evidence card?

A. Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. B. A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. C. The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. D. The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next.

**Answer: B. Explanation: A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q3. Which statement uses Minimum prize evidence card without changing its institution, unit or status?

A. Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. B. The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. C. A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. D. Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution.

**Answer: C. Explanation: A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q4. Which option avoids the standard UPSC close-option trap about Minimum prize evidence card?

A. The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. B. Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. C. A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. D. A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field.

**Answer: D. Explanation: A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q5. Which statement correctly identifies Discovery invention theory and observation?

A. A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. B. The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. C. The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. D. Birthplace, citizenship or nationality,

education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution.

**Answer: A. Explanation: A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q6. Which option preserves the technical boundary of Discovery invention theory and observation?**

A. The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. B. A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. C. Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. D. Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution.

**Answer: B. Explanation: A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q7. Which statement uses Discovery invention theory and observation without changing its institution, unit or status?**

A. The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. B. Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. C. A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. D. A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim.

**Answer: C. Explanation: A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**



### Q8. Which option avoids the standard UPSC close-option trap about Discovery invention theory and observation?

A. Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. B. The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. C. A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. D. A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly.

**Answer: D. Explanation: A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q9. Which statement correctly identifies Nobel category architecture?

A. The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. B. Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. C. Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. D. The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next.

**Answer: A. Explanation: The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q10. Which option preserves the technical boundary of Nobel category architecture?

A. A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. B. The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. C. Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit

should follow the specific official citation and named contribution. D. The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next.

**Answer: B. Explanation: The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q11. Which statement uses Nobel category architecture without changing its institution, unit or status?**

A. The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. B. A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. C. The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. D. Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution.

**Answer: C. Explanation: The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q12. Which option avoids the standard UPSC close-option trap about Nobel category architecture?**

A. The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. B. A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. C. The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. D. The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field.

**Answer: D. Explanation: The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field.**

The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

**Q13. Which statement correctly identifies Laureate institution nationality boundary?**

A. Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. B. Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. C. A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. D. The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next.

**Answer: A. Explanation: Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q14. Which option preserves the technical boundary of Laureate institution nationality boundary?**

A. Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. B. Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. C. A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. D. The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing.

**Answer: B. Explanation: Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q15. Which statement uses Laureate institution nationality boundary without changing its institution, unit or status?**

A. The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. B. The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. C. Birthplace, citizenship or nationality, education, research location and affiliation at

the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. D. A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim.

**Answer: C. Explanation: Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### **Q16. Which option avoids the standard UPSC close-option trap about Laureate institution nationality boundary?**

A. The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. B. The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. C. The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. D. Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution.

**Answer: D. Explanation: Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### **Q17. Which statement correctly identifies Citation-to-policy evidence ladder?**

A. The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. B. A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. C. Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. D. The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing.

**Answer: A. Explanation: The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q18. Which option preserves the technical boundary of Citation-to-policy evidence ladder?

A. A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. B. The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. C. The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. D. The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer.

**Answer: B. Explanation: The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q19. Which statement uses Citation-to-policy evidence ladder without changing its institution, unit or status?

A. The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. B. The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. C. The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. D. The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality.

**Answer: C. Explanation: The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q20. Which option avoids the standard UPSC close-option trap about Citation-to-policy evidence ladder?

A. The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. B. The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. C. The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. D. The official citation identifies



the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next.

**Answer: D. Explanation: The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### **Q21. Which statement correctly identifies Chronology and attribution discipline?**

A. Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. B. The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. C. The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. D. A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim.

**Answer: A. Explanation: Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### **Q22. Which option preserves the technical boundary of Chronology and attribution discipline?**

A. The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. B. Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. C. The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. D. The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer.

**Answer: B. Explanation: Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q23. Which statement uses Chronology and attribution discipline without changing its institution, unit or status?**

A. The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. B. The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. C. Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. D. The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality.

**Answer: C. Explanation: Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q24. Which option avoids the standard UPSC close-option trap about Chronology and attribution discipline?**

A. The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. B. The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. C. Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. D. Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution.

**Answer: D. Explanation: Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q25. Which statement correctly identifies Replication uncertainty and mature evidence?**

A. A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. B. The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. C. The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. D. The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley.

Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality.

**Answer: A. Explanation: A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q26. Which option preserves the technical boundary of Replication uncertainty and mature evidence?**

A. The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. B. A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. C. The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. D. The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer.

**Answer: B. Explanation: A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q27. Which statement uses Replication uncertainty and mature evidence without changing its institution, unit or status?**

A. The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. B. The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. C. A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. D. Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment.

**Answer: C. Explanation: A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**



### Q28. Which option avoids the standard UPSC close-option trap about Replication uncertainty and mature evidence?

A. The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. B. Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. C. The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. D. A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim.

**Answer: D. Explanation: A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q29. Which statement correctly identifies Physics 2025 award citation?

A. The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. B. The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. C. The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. D. The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country.

**Answer: A. Explanation: The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q30. Which option preserves the technical boundary of Physics 2025 award citation?

A. The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. B. The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. C. Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. D. The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country.

**Answer: B. Explanation: The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q31. Which statement uses Physics 2025 award citation without changing its institution, unit or status?**

A. Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. B. The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. C. The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. D. The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country.

**Answer: C. Explanation: The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q32. Which option avoids the standard UPSC close-option trap about Physics 2025 award citation?**

A. Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. B. Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. C. The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. D. The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing.

**Answer: D. Explanation: The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q33. Which statement correctly identifies Physics 2025 mechanism?

A. The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. B. The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. C. The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. D. Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment.

**Answer: A. Explanation: The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q34. Which option preserves the technical boundary of Physics 2025 mechanism?

A. Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. B. The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. C. The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. D. The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues.

**Answer: B. Explanation: The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q35. Which statement uses Physics 2025 mechanism without changing its institution, unit or status?

A. The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. B. Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. C. The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. D. Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell

development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome.

**Answer: C. Explanation: The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### **Q36. Which option avoids the standard UPSC close-option trap about Physics 2025 mechanism?**

A. Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. B. The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. C. The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. D. The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer.

**Answer: D. Explanation: The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### **Q37. Which statement correctly identifies John Clarke biographical route?**

A. The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. B. Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. C. The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. D. The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country.

**Answer: A. Explanation: The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q38. Which option preserves the technical boundary of John Clarke biographical route?

A. Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. B. The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. C. Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. D. The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues.

**Answer: B. Explanation: The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q39. Which statement uses John Clarke biographical route without changing its institution, unit or status?

A. Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. B. The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. C. The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. D. The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues.

**Answer: C. Explanation: The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q40. Which option avoids the standard UPSC close-option trap about John Clarke biographical route?

A. C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. B. Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. C. The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary



messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. D. The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality.

**Answer: D. Explanation: The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### **Q41. Which statement correctly identifies Chemistry 2025 award citation?**

A. The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. B. Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. C. The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. D. Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome.

**Answer: A. Explanation: The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### **Q42. Which option preserves the technical boundary of Chemistry 2025 award citation?**

A. Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. B. The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. C. The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. D. The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues.

**Answer: B. Explanation: The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**



**Q43. Which statement uses Chemistry 2025 award citation without changing its institution, unit or status?**

A. Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. B. C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. C. The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. D. The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy.

**Answer: C. Explanation: The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q44. Which option avoids the standard UPSC close-option trap about Chemistry 2025 award citation?**

A. The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. B. Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. C. C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. D. The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country.

**Answer: D. Explanation: The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q45. Which statement correctly identifies Metal-organic framework mechanism?**

A. Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. B. Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. C. The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. D. The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E.

Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues.

**Answer: A. Explanation: Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q46. Which option preserves the technical boundary of Metal-organic framework mechanism?**

A. C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. B. Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. C. Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. D. The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy.

**Answer: B. Explanation: Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q47. Which statement uses Metal-organic framework mechanism without changing its institution, unit or status?**

A. C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. B. Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. C. Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. D. The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy.

**Answer: C. Explanation: Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. The other options change the scientific category, institutional owner, measurement, mission**

**Q48. Which option avoids the standard UPSC close-option trap about Metal-organic framework mechanism?**

A. Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself. B. Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. C. C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. D. Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment.

**Answer: D. Explanation: Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q49. Which statement correctly identifies Medicine 2025 award citation?**

A. The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. B. Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. C. C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. D. The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy.

**Answer: A. Explanation: The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q50. Which option preserves the technical boundary of Medicine 2025 award citation?

A. The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. B. The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. C. C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. D. Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids.

**Answer: B. Explanation: The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q51. Which statement uses Medicine 2025 award citation without changing its institution, unit or status?

A. C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. B. Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. C. The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. D. Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself.

**Answer: C. Explanation: The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q52. Which option avoids the standard UPSC close-option trap about Medicine 2025 award citation?

A. Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. B. The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. C. Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later

application rather than the citation itself. D. The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues.

**Answer: D. Explanation: The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q53. Which statement correctly identifies Peripheral tolerance mechanism?

A. Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. B. C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. C. The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. D. Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids.

**Answer: A. Explanation: Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q54. Which option preserves the technical boundary of Peripheral tolerance mechanism?

A. Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself. B. Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. C. Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. D. C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation.

**Answer: B. Explanation: Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**



**Q55. Which statement uses Peripheral tolerance mechanism without changing its institution, unit or status?**

A. The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. B. Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. C. Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. D. Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself.

**Answer: C. Explanation: Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q56. Which option avoids the standard UPSC close-option trap about Peripheral tolerance mechanism?**

A. Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself. B. Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. C. The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. D. Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome.

**Answer: D. Explanation: Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q57. Which statement correctly identifies Medicine 2024 microRNA anchor?**

A. The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. B. C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. C. Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of



Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself. D. Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids.

**Answer: A. Explanation: The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q58. Which option preserves the technical boundary of Medicine 2024 microRNA anchor?**

A. Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself. B. The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. C. Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. D. The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation.

**Answer: B. Explanation: The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q59. Which statement uses Medicine 2024 microRNA anchor without changing its institution, unit or status?**

A. The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. B. Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. C. The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. D. Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself.

**Answer: C. Explanation: The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q60. Which option avoids the standard UPSC close-option trap about Medicine 2024 microRNA anchor?

A. The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. B. Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. C. A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. D. The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy.

**Answer: D. Explanation: The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q61. Which statement correctly identifies Raman prize and mechanism?

A. C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. B. The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. C. Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. D. Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself.

**Answer: A. Explanation: C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q62. Which option preserves the technical boundary of Raman prize and mechanism?

A. Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. B. C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. C. Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself. D.

The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation.

**Answer: B. Explanation: C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q63. Which statement uses Raman prize and mechanism without changing its institution, unit or status?**

A. Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. B. A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. C. C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. D. The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation.

**Answer: C. Explanation: C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q64. Which option avoids the standard UPSC close-option trap about Raman prize and mechanism?**

A. A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. B. A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. C. Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. D. C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation.

**Answer: D. Explanation: C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q65. Which statement correctly identifies Khorana genetic-code prize?

A. Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. B. The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. C. Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself. D. Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates.

**Answer: A. Explanation: Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q66. Which option preserves the technical boundary of Khorana genetic-code prize?

A. Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. B. Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. C. The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. D. A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field.

**Answer: B. Explanation: Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q67. Which statement uses Khorana genetic-code prize without changing its institution, unit or status?

A. Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. B. A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. C. Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. D. A theory explains and predicts, an observation records a phenomenon, an experiment

tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly.

**Answer: C. Explanation: Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q68. Which option avoids the standard UPSC close-option trap about Khorana genetic-code prize?**

A. A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. B. The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. C. A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. D. Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids.

**Answer: D. Explanation: Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q69. Which statement correctly identifies Ramakrishnan ribosome prize?**

A. Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself. B. The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. C. A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. D. Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates.

**Answer: A. Explanation: Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**



### Q70. Which option preserves the technical boundary of Ramakrishnan ribosome prize?

A. A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. B. Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself. C. A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. D. Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates.

**Answer: B. Explanation: Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q71. Which statement uses Ramakrishnan ribosome prize without changing its institution, unit or status?

A. A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. B. A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. C. Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself. D. The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field.

**Answer: C. Explanation: Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q72. Which option avoids the standard UPSC close-option trap about Ramakrishnan ribosome prize?

A. A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. B. The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. C. Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or

author of every contribution. D. Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself.

**Answer: D. Explanation: Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q73. Which statement correctly identifies Routed PYQ mechanism-first use?

A. The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. B. A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. C. Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. D. A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field.

**Answer: A. Explanation: The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

### Q74. Which option preserves the technical boundary of Routed PYQ mechanism-first use?

A. A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. B. The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. C. A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. D. The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field.

**Answer: B. Explanation: The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q75. Which statement uses Routed PYQ mechanism-first use without changing its institution, unit or status?**

A. Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. B. The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. C. The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. D. A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly.

**Answer: C. Explanation: The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q76. Which option avoids the standard UPSC close-option trap about Routed PYQ mechanism-first use?**

A. The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. B. Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. C. The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. D. The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation.

**Answer: D. Explanation: The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q77. Which statement correctly identifies Prize claim and current-status firewall?**

A. Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. B. The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. C. A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered

field, employer or popular headline cannot replace any missing field. D. A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly.

**Answer: A. Explanation: Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### **Q78. Which option preserves the technical boundary of Prize claim and current-status firewall?**

A. A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. B. Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. C. The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. D. Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution.

**Answer: B. Explanation: Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

#### **Q79. Which statement uses Prize claim and current-status firewall without changing its institution, unit or status?**

A. Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. B. The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. C. Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. D. The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next.

**Answer: C. Explanation: Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Q80. Which option avoids the standard UPSC close-option trap about Prize claim and current-status firewall?**

A. The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. B. Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. C. Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. D. Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates.

**Answer: D. Explanation: Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.**

**Semantic-completeness coverage drills - Topic 26**



| Drill | Prompt                                                                                                                                                                                                                                                                                                                                                                                | Minimum answer route                                                               | Fatal trap                   |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------|
| A     | The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next.            |                                                                                    |                              |
| B     | Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution.                             |                                                                                    |                              |
| C     | Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution.                                           |                                                                                    |                              |
| D     | Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. |                                                                                    |                              |
| E     | How did temples and monasteries matter?                                                                                                                                                                                                                                                                                                                                               | ritual + land + labour + redistribution + archive + local variation                | merely religious             |
| F     | Was Brahmanization passive assimilation?                                                                                                                                                                                                                                                                                                                                              | grant/cult/status -> localization -> tribal/local agency -> unequal reciprocity    | one-way replacement          |
| G     | Evaluate social change.                                                                                                                                                                                                                                                                                                                                                               | varna text -> jati/occupation/exclusion -> gender/property/labour -> source limits | prescription equals practice |
| H     | Compare regions.                                                                                                                                                                                                                                                                                                                                                                      | north + east + Deccan + south/frontier -> different sequences                      | one pan-Indian model         |

**PYQ status drill:** all ten retained PYQs are adjacent-owned; none is routed directly to Topic 26.

## PYQS AND ANSWER PRACTICE

### AUDITED TOPIC-SPECIFIC PYQ OWNERSHIP

The audited ledgers route 2026 Prelims Q81 directly to this owner. The 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain directly owned by Topic 21 and appear only as cross-owner examples of attribution and citation-to-application discipline. Three representative cards preserve that boundary and reproduce no objective answer.

## PYQ DEMAND CARD 1 - 2026 Prelims GS-I

**Demand:** Identify the 2025 Nobel Prize recipient by matching biographical and professional clues to the official category, citation and award-time affiliation.

**Status:** Verified direct routed objective demand covering 2026 Q81; the locally held Set-A key is provisional, so no option, answer letter or inferred key is supplied.

**Model solution: Minimum prize evidence card:** A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. **Laureate institution nationality boundary:** Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. **Physics 2025 award citation:** The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. **John Clarke biographical route:** The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. **Routed PYQ mechanism-first use:** The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

**Demand decoding:** Treat "PYQ DEMAND CARD 1 - 2026 Prelims GS-I" as a mechanism, category, institution, unit, date and status problem.

### Detailed examiner-grade model answer:

**Introduction and thesis: Minimum prize evidence card:** A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. **Laureate institution nationality boundary:** Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. **Physics 2025 award citation:** The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. **John Clarke biographical route:** The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. **Routed PYQ mechanism-first use:** The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

### Analytical body:

- Claim and named evidence:** Demand: Identify the 2025 Nobel Prize recipient by matching biographical and professional clues to the official category, citation and award-time affiliation. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
- Claim and named evidence:** Status: Verified direct routed objective demand covering 2026 Q81; the locally held Set-A key is provisional, so no option, answer letter or inferred key is supplied. **Analysis:** Connect mechanism ->

named system/institution -> application or implementation pathway -> public or strategic consequence.

**Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

**Counter-position / limit:** A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

**Qualified conclusion: Minimum prize evidence card:** A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. **Laureate institution nationality boundary:** Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. **Physics 2025 award citation:** The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. **John Clarke biographical route:** The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. **Routed PYQ mechanism-first use:** The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

**Executable exam-length answer / compression plan:** Identify the object and actor; trace the technical mechanism; fix the measurement and platform; test each statement against source date, status rung and closest exception.

**Why this earns marks:** It prevents organisation, platform, process, application, regulatory role, target and observed outcome from being conflated.

**How to improve this answer:** For "PYQ DEMAND CARD 1 - 2026 Prelims GS-I", explain why the nearest distractor fails on mechanism, platform, unit, institution, legal status, maturity or date.

## PYQ DEMAND CARD 2 - 2018 GS-III

**Demand:** Use citation and discovery-attribution discipline while discussing the contribution of Bose-Einstein statistics to physics.

**Status:** Cross-owner supporting card: Topic 21 retains the substantive statistics and physics answer; Topic 26 contributes attribution, chronology and discovery-to-application boundaries without claiming a Nobel citation for the routed concept.

**Model solution: Discovery invention theory and observation:** A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. **Citation-to-policy evidence ladder:** The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. **Chronology and attribution discipline:** Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. **Routed PYQ mechanism-first use:** The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. The solution therefore answers the routed demand without

inventing an official model answer or an objective answer key.

**Demand decoding:** The directive **answer** requires a direct position on "PYQ DEMAND CARD 2 - 2018 GS-III", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

**Detailed examiner-grade model answer:**

**Introduction and thesis: Discovery invention theory and observation:** A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. **Citation-to-policy evidence ladder:** The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. **Chronology and attribution discipline:** Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. **Routed PYQ mechanism-first use:** The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

**Analytical body:**

1. **Claim and named evidence:** Demand: Use citation and discovery-attribution discipline while discussing the contribution of Bose-Einstein statistics to physics. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

**Counter-position / limit:** A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

**Qualified conclusion: Discovery invention theory and observation:** A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. **Citation-to-policy evidence ladder:** The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. **Chronology and attribution discipline:** Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. **Routed PYQ mechanism-first use:** The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

**Executable exam-length answer / compression plan:** For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

**How to improve this answer:** For "PYQ DEMAND CARD 2 - 2018 GS-III", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation

constraint and answer-specific qualification.

### PYQ DEMAND CARD 3 - 2021 GS-III

**Demand:** Explain the everyday-life impact of the blue LED invention recognised by the 2014 Nobel Prize while separating award citation, mechanism, application and outcome.

**Status:** Cross-owner supporting card: Topic 21 retains semiconductor and LED physics; Topic 26 supplies prize-citation, invention, application and commercialization discipline rather than an official model answer.

**Model solution: Discovery invention theory and observation:** A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. **Citation-to-policy evidence ladder:** The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. **Chronology and attribution discipline:** Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. **Replication uncertainty and mature evidence:** A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. **Routed PYQ mechanism-first use:** The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

**Demand decoding:** The directive **answer** requires a direct position on "PYQ DEMAND CARD 3 - 2021 GS-III", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

#### Detailed examiner-grade model answer:

**Introduction and thesis: Discovery invention theory and observation:** A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. **Citation-to-policy evidence ladder:** The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. **Chronology and attribution discipline:** Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. **Replication uncertainty and mature evidence:** A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. **Routed PYQ mechanism-first use:** The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

#### Analytical body:

1. **Claim and named evidence:** Demand: Explain the everyday-life impact of the blue LED invention recognised by the 2014 Nobel Prize while separating award citation, mechanism, application and outcome. **Analysis:** Connect



mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

2. **Claim and named evidence:** Status: Cross-owner supporting card: Topic 21 retains semiconductor and LED physics; Topic 26 supplies prize-citation, invention, application and commercialization discipline rather than an official model answer. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

**Counter-position / limit:** A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

**Qualified conclusion: Discovery invention theory and observation:** A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. **Citation-to-policy evidence ladder:** The official citation identifies the awarded contribution; the underlying discovery explains the mechanism; a possible application shows technical relevance; engineering validation tests reproducibility and performance; commercialization requires production and adoption; and a policy outcome requires measured social effect. None of these rungs proves the next. **Chronology and attribution discipline:** Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. **Replication uncertainty and mature evidence:** A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. **Routed PYQ mechanism-first use:** The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

**Executable exam-length answer / compression plan:** For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

**How to improve this answer:** For "PYQ DEMAND CARD 3 - 2021 GS-III", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

## ORIGINAL MAINS 1 - 10 MARKS

**Question:** Explain the minimum evidence needed to identify a Nobel laureate and distinguish biographical clues. Answer in about 150 words.

**Model thesis: Claim:** Minimum prize evidence card. **Named evidence/example:** A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Nobel category architecture. **Named evidence/example:** The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges

Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Laureate institution nationality boundary. **Named evidence/example:** Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim -> named evidence -> analysis -> qualification:**

- A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field.
- The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field.
- Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution.

**Qualified conclusion: Claim:** Minimum prize evidence card. **Named evidence/example:** A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Nobel category architecture. **Named evidence/example:** The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Laureate institution nationality boundary. **Named evidence/example:** Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Demand decoding:** The directive **explain** requires a direct position on "Explain the minimum evidence needed to identify a Nobel laureate and distinguish biographical...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

**Detailed examiner-grade model answer:**

**Introduction and thesis: Claim:** Minimum prize evidence card. **Named evidence/example:** A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

**Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Nobel category architecture. **Named evidence/example:** The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Laureate institution nationality boundary. **Named evidence/example:** Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

#### Analytical body:

1. **Claim and named evidence:** A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** Birthplace, citizenship or nationality, education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

**Counter-position / limit:** A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

**Qualified conclusion:** **Claim:** Minimum prize evidence card. **Named evidence/example:** A reliable prize card records exact prize category and year, official laureate spelling, official motivation or citation, plain-language mechanism, birthplace only when stated, award-time affiliation, direct India link if sourced and verification date. A remembered field, employer or popular headline cannot replace any missing field. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Nobel category architecture. **Named evidence/example:** The Nobel categories connect Physics, Chemistry, Physiology or Medicine, Literature and Peace to Alfred Nobel's original framework, while the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel was established later. Awarding institutions differ, and Peace can recognise organisations; the word Nobel alone does not establish category, recipient type or scientific field. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Laureate institution nationality boundary. **Named evidence/example:** Birthplace, citizenship or nationality,

education, research location and affiliation at the time of award are different biographical facts. A prize can be shared among laureates at different institutions, and an institution supplies a research environment without becoming an automatic co-recipient or author of every contribution. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Executable exam-length answer / compression plan:** For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

**How to improve this answer:** For "Explain the minimum evidence needed to identify a Nobel laureate and distinguish biographical...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

## ORIGINAL MAINS 2 - 10 MARKS

**Question:** Distinguish theory, observation, experiment, discovery, invention and development in prize attribution. Answer in about 150 words.

**Model thesis: Claim:** Discovery invention theory and observation. **Named evidence/example:** A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Chronology and attribution discipline. **Named evidence/example:** Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Replication uncertainty and mature evidence. **Named evidence/example:** A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim -> named evidence -> analysis -> qualification:**

- A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly.
- Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution.
- A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which



institution verified each later claim.

**Qualified conclusion: Claim:** Discovery invention theory and observation. **Named evidence/example:** A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Chronology and attribution discipline. **Named evidence/example:** Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Replication uncertainty and mature evidence. **Named evidence/example:** A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Demand decoding:** The directive **answer** requires a direct position on "Distinguish theory, observation, experiment, discovery, invention and development in prize...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

#### **Detailed examiner-grade model answer:**

**Introduction and thesis: Claim:** Discovery invention theory and observation. **Named evidence/example:** A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Chronology and attribution discipline. **Named evidence/example:** Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Replication uncertainty and mature evidence. **Named evidence/example:** A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

#### **Analytical body:**

1. **Claim and named evidence:** A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit,



source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

2. **Claim and named evidence:** Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

**Counter-position / limit:** A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

**Qualified conclusion:** **Claim:** Discovery invention theory and observation. **Named evidence/example:** A theory explains and predicts, an observation records a phenomenon, an experiment tests under controlled conditions, a discovery establishes a phenomenon or relation, and an invention creates a device or process. A Nobel citation may recognise a discovery, development or method, so the awarded verb and object must be preserved exactly. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Chronology and attribution discipline. **Named evidence/example:** Scientific progress often spans prior theory, enabling instruments, decisive experiments, independent replication and later engineering by many teams. Prize year is the award year rather than automatically the discovery, publication, patent, product or deployment year; credit should follow the specific official citation and named contribution. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Replication uncertainty and mature evidence. **Named evidence/example:** A single result, preprint, company announcement or prize does not eliminate measurement error, boundary conditions or scaling constraints. Strong scientific reasoning asks whether the effect was independently reproduced, whether alternative explanations were tested, what population or operating range applies and which institution verified each later claim. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Executable exam-length answer / compression plan:** For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

**How to improve this answer:** For "Distinguish theory, observation, experiment, discovery, invention and development in prize...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

## ORIGINAL MAINS 3 - 15 MARKS

**Question:** Analyse the 2025 Physics prize through its official citation, superconducting-circuit mechanism and technology boundary. Answer in about 250 words.

**Model thesis: Claim:** Physics 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Physics 2025 mechanism. **Named evidence/example:** The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** John Clarke biographical route. **Named evidence/example:** The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim -> named evidence -> analysis -> qualification:**

- The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing.
- The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer.
- The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality.

**Qualified conclusion: Claim:** Physics 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Physics 2025 mechanism. **Named evidence/example:** The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** John Clarke biographical route. **Named evidence/example:** The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion

retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Demand decoding:** The directive **analyse** requires a direct position on "Analyse the 2025 Physics prize through its official citation, superconducting-circuit...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

**Detailed examiner-grade model answer:**

**Introduction and thesis: Claim:** Physics 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

**Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Physics 2025 mechanism. **Named evidence/example:** The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim:** John Clarke biographical route. **Named evidence/example:** The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Analytical body:**

1. **Claim and named evidence:** The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

**Counter-position / limit:** A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

**Qualified conclusion: Claim:** Physics 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic

quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

**Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Physics 2025 mechanism. **Named evidence/example:** The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim:** John Clarke biographical route. **Named evidence/example:** The official facts page records John Clarke as born in Cambridge, United Kingdom, in 1942 and affiliated at the time of the award with the University of California, Berkeley. Birthplace and United States institutional affiliation are separate clues, while his one-third prize share identifies neither sole authorship nor nationality. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Executable exam-length answer / compression plan:** For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

**How to improve this answer:** For "Analyse the 2025 Physics prize through its official citation, superconducting-circuit...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

## ORIGINAL MAINS 4 - 15 MARKS

**Question:** Discuss the 2025 Chemistry and Medicine prizes by separating awarded discoveries from possible applications and commercialization. Answer in about 250 words.

**Model thesis: Claim:** Chemistry 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim:** Metal-organic framework mechanism. **Named evidence/example:** Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Medicine 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues.

**Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Peripheral tolerance mechanism. **Named evidence/example:** Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked



autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim -> named evidence -> analysis -> qualification:**

- The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country.
- Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment.
- The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues.
- Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome.

**Qualified conclusion: Claim:** Chemistry 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim:** Metal-organic framework mechanism. **Named evidence/example:** Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Medicine 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Peripheral tolerance mechanism. **Named evidence/example:** Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Demand decoding:** The directive **discuss** requires a direct position on "Discuss the 2025 Chemistry and Medicine prizes by separating awarded discoveries from...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

**Detailed examiner-grade model answer:**



**Introduction and thesis:** **Claim:** Chemistry 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim:** Metal-organic framework mechanism. **Named evidence/example:** Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim:** Medicine 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim:** Peripheral tolerance mechanism. **Named evidence/example:** Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

#### Analytical body:

1. **Claim and named evidence:** The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

exception or residual risk.

**Counter-position / limit:** A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

**Qualified conclusion: Claim:** Chemistry 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim:** Metal-organic framework mechanism. **Named evidence/example:** Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Medicine 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Peripheral tolerance mechanism. **Named evidence/example:** Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Executable exam-length answer / compression plan:** For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

**How to improve this answer:** For "Discuss the 2025 Chemistry and Medicine prizes by separating awarded discoveries from...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

## ORIGINAL MAINS 5 - 20 MARKS

**Question:** Examine how recent Nobel anchors in quantum physics, metal-organic frameworks, immune tolerance and microRNA reveal the path from basic science to application. Answer in about 300 words.

**Model thesis: Claim:** Physics 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Physics 2025 mechanism. **Named evidence/example:** The laureates used a

superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Chemistry 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Metal-organic framework mechanism. **Named evidence/example:** Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Medicine 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Peripheral tolerance mechanism. **Named evidence/example:** Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Medicine 2024 microRNA anchor. **Named evidence/example:** The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim -> named evidence -> analysis -> qualification:**

- The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing.
- The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer.
- The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country.
- Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but

capability does not prove scale, lifecycle benefit or commercial deployment.

- The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues.
- Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome.
- The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy.

**Qualified conclusion: Claim:** Physics 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

**Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Physics 2025 mechanism. **Named evidence/example:** The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim:** Chemistry 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country.

**Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Metal-organic framework mechanism. **Named evidence/example:** Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment.

**Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Medicine 2025 award citation.

**Named evidence/example:** The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Peripheral tolerance mechanism. **Named**

**evidence/example:** Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Medicine 2024 microRNA anchor. **Named evidence/example:** The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger



RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Demand decoding:** The directive **examine** requires a direct position on "Examine how recent Nobel anchors in quantum physics, metal-organic frameworks, immune...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

#### **Detailed examiner-grade model answer:**

**Introduction and thesis: Claim:** Physics 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

**Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Physics 2025 mechanism. **Named evidence/example:** The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim:** Chemistry 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country.

**Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Metal-organic framework mechanism. **Named evidence/example:** Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment.

**Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Medicine 2025 award citation.

**Named evidence/example:** The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

**Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Peripheral tolerance mechanism. **Named**

**evidence/example:** Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Medicine 2024 microRNA anchor. **Named evidence/example:** The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution,



system category, legal character and development-to-deployment status boundary.

#### Analytical body:

1. **Claim and named evidence:** The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
5. **Claim and named evidence:** The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance. The contribution concerns how the immune system is kept from attacking the body's own tissues. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
6. **Claim and named evidence:** Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
7. **Claim and named evidence:** The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

**Counter-position / limit:** A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

**Qualified conclusion: Claim:** Physics 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Physics was awarded to John Clarke, Michel H. Devoret and John M. Martinis for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. The official award date was 7 October 2025, and the citation is narrower than the broad label quantum computing. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

**Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Physics 2025 mechanism. **Named evidence/example:** The laureates used a superconducting electrical circuit with a Josephson junction and showed a collective macroscopic state escaping a zero-voltage condition through quantum tunnelling and absorbing or emitting specific energy amounts. The experiment demonstrated macroscopic quantum behaviour; it did not itself constitute a fault-tolerant general-purpose quantum computer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim:** Chemistry 2025 award citation. **Named evidence/example:** The 2025 Nobel Prize in Chemistry was awarded on 8 October 2025 to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for the development of metal-organic frameworks. The citation concerns a material architecture, not one environmental policy, company, product or country.

**Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim:** Metal-organic framework mechanism. **Named evidence/example:** Metal-organic frameworks use metal ions as nodes linked by organic molecules into porous crystalline structures with cavities; changing building blocks can tune capture, storage, transport, catalysis or conductivity. Official examples include water harvesting, carbon-dioxide capture, toxic-gas storage and chemical reactions, but capability does not prove scale, lifecycle benefit or commercial deployment.

**Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Medicine 2025 award citation.

**Named evidence/example:** The 2025 Nobel Prize in Physiology or Medicine was awarded on 6 October 2025 to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance.

The contribution concerns how the immune system is kept from attacking the body's own tissues. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim:** Peripheral tolerance mechanism. **Named evidence/example:** Shimon Sakaguchi identified regulatory T cells, while Brunkow and Ramsdell linked autoimmune vulnerability to Foxp3 and Sakaguchi connected Foxp3 to regulatory-T-cell development. These discoveries explain a control layer beyond central tolerance; references to cancer, autoimmunity or transplantation research and clinical trials do not prove approved treatment or population outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim:** Medicine 2024 microRNA anchor. **Named evidence/example:** The 2024 Nobel Prize in Physiology or Medicine jointly recognised Victor Ambros and Gary Ruvkun for discovery of microRNA and its role in post-transcriptional gene regulation. The mechanism shows that small RNAs can bind complementary messenger RNA and inhibit protein production or promote RNA degradation; it does not turn every RNA association into a therapy. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Executable exam-length answer / compression plan:** For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

**How to improve this answer:** For "Examine how recent Nobel anchors in quantum physics, metal-organic frameworks, immune...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

## ORIGINAL MAINS 6 - 20 MARKS

**Question:** Evaluate discovery attribution and Indian scientific relevance through Raman, Khorana, Ramakrishnan and routed PYQs. Answer in about 300 words.

**Model thesis: Claim:** Raman prize and mechanism. **Named evidence/example:** C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Khorana genetic-code prize. **Named evidence/example:** Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Ramakrishnan ribosome prize. **Named evidence/example:** Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Routed PYQ mechanism-first use. **Named evidence/example:** The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Prize claim and current-status firewall. **Named evidence/example:** Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Claim -> named evidence -> analysis -> qualification:**

- C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation.
- Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids.
- Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and

Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself.

- The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation.
- Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates.

**Qualified conclusion: Claim:** Raman prize and mechanism. **Named evidence/example:** C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Khorana genetic-code prize. **Named evidence/example:** Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Ramakrishnan ribosome prize. **Named evidence/example:** Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Routed PYQ mechanism-first use. **Named evidence/example:** The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Prize claim and current-status firewall. **Named evidence/example:** Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

**Demand decoding:** The directive **evaluate** requires a direct position on "Evaluate discovery attribution and Indian scientific relevance through Raman, Khorana,...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

#### **Detailed examiner-grade model answer:**

**Introduction and thesis: Claim:** Raman prize and mechanism. **Named evidence/example:** C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit,



institution, system category, legal character and development-to-deployment status boundary. **Claim:** Khorana genetic-code prize. **Named evidence/example:** Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Ramakrishnan ribosome prize. **Named evidence/example:** Venkatraman Ramakrishnan shared the 2009 Nobel Prize in Chemistry for studies of the structure and function of the ribosome, with the MRC Laboratory of Molecular Biology in Cambridge as award-time affiliation and Chidambaram, Tamil Nadu, India, as birthplace. X-ray crystallography mapped ribosomal structure; antibiotic relevance is a later application rather than the citation itself. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Routed PYQ mechanism-first use. **Named evidence/example:** The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Prize claim and current-status firewall. **Named evidence/example:** Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

#### Analytical body:

1. **Claim and named evidence:** C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
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4. **Claim and named evidence:** The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. **Analysis:** Connect mechanism -> named system/institution -> application or



implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

5. **Claim and named evidence:** Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence.

**Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

**Counter-position / limit:** A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

**Qualified conclusion: Claim:** Raman prize and mechanism. **Named evidence/example:** C. V. Raman received the 1930 Nobel Prize in Physics for work on scattering of light and discovery of the effect named after him, with Calcutta University as award-time affiliation. The Raman effect involves a small fraction of scattered light changing wavelength through energy exchange with molecules and supports material analysis; later use is distinct from the original citation.

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**Named evidence/example:** Har Gobind Khorana shared the 1968 Nobel Prize in Physiology or Medicine for interpretation of the genetic code and its function in protein synthesis, with the University of Wisconsin as award-time affiliation and Raipur, India, as birthplace on the official page. His experimental contribution used constructed RNA chains to help map codons to amino acids. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

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**Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Routed PYQ mechanism-first use.

**Named evidence/example:** The direct 2026 routed demand identifies a 2025 laureate from biographical and professional clues, while the 2018 Bose-Einstein-statistics and 2021 blue-LED Mains demands remain with Topic 21. Topic 26 supplies citation, attribution and discovery-to-application discipline but must not displace the specialist physics explanation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Prize claim and current-status firewall. **Named evidence/example:** Nomination, announcement, award citation, laureate biography, scientific mechanism, possible application, clinical trial, patent, licensed product, manufacturing scale, market adoption and policy outcome are distinct statuses. The 2026 prizes had not been announced by 4 September 2026, so recent anchors stop at official 2025 awards rather than inventing future laureates. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

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**Executable exam-length answer / compression plan:** For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

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